

Plan 3 Graph-Reduction Recheck

May 2026

1 Purpose

I rechecked the Plan 3 material for the specific claim that the reduction from a good torsion level to a radial graph is already closed. The conclusion is negative in the literal sense: the repo contains several real ingredients for such a reduction, but no complete proof that the stopped level is, or can be repaired at $O(\delta_T)$ cost into, a small L^∞ radial graph.

The closest-looking closures are:

1. `C-assembled-proof-steps/proof-step2.md`, especially Sections 7–9, plus `B-routeB-weinberger/verify-`
2. `C-assembled-proof-steps/proof-step4.md`, which explains how an L^∞ Serrin output would enter the graph theorem;
3. `A-strategyA-graph/expA-anygraph-monotonicity.md` and the Strategy B notes;
4. `B-routeB-weinberger/tentacle-model.md`;
5. the Plan 2 metric/interval route, which replaces graph coordinates by an L^1 -modulo-translations metric trace.

None of these proves the desired graph-entry or radialization theorem as written.

2 The strongest apparent closure: Chain B

The text in `proof-step2.md` says that the graph substatement is unnecessary. It proposes the chain

$$\delta_T(D) \gtrsim |D|^{-1} \int_D \left| D^2 u_D + \frac{1}{n} I \right|^2 \implies \mathcal{A}(D)^2 \lesssim \int_D \left| D^2 u_D + \frac{1}{n} I \right|^2,$$

for $D = E_{\hat{\Gamma}}$. This would indeed bypass graph entry. The problem is that the first inequality is false, even on smooth volume-normalized near-balls.

Let Y_k be an $L^2(\mathbb{S}^{n-1})$ -normalized spherical harmonic of degree $k \geq 2$ and

$$D_{\varepsilon,k} = \{r < 1 + \varepsilon Y_k + O_k(\varepsilon^2)\},$$

with the $O_k(\varepsilon^2)$ constant mode enforcing volume. The elementary ball linearization gives

$$\delta_T(D_{\varepsilon,k}) = \frac{\varepsilon^2}{2n^2}(k-1) + O_k(\varepsilon^3),$$

whereas

$$\int_{D_{\varepsilon,k}} \left| D^2 u_{D_{\varepsilon,k}} + \frac{1}{n} I \right|^2 = \frac{\varepsilon^2}{n^2} (k-1)k(2k+n-2) + O_k(\varepsilon^3).$$

Thus

$$\frac{\delta_T(D_{\varepsilon,k})}{\int_{D_{\varepsilon,k}} |D^2 u_{D_{\varepsilon,k}} + I/n|^2} \rightarrow \frac{1}{2k(2k+n-2)} \rightarrow 0.$$

This is the calculation recorded in `PLAN3_N2_SINGLE_LEVEL_RECHECK.tex`. It kills the unweighted Hessian coercivity used by Chain B. Since the counterexamples are smooth small radial graphs, the failure is not caused by tentacles, disconnectedness, or rough boundary.

3 Why the auxiliary H^2 -to-geometry text does not repair this

Even if a small unweighted H^2 defect were somehow available, the conversion argument in `proof-step2.md` Section 7 is not a proved graph-reduction theorem.

The proof asserts that after removing a bad set Z , the remaining domain is a uniformly John/Lipschitz union foliated by nondegenerate levels, and that local harmonic estimates can be chained to one constant vector. The inputs cited there are:

$$\mathcal{H}^{n-1}\{|\nabla u| \leq s\} \lesssim D_H s$$

and a model tube estimate saying that the unweighted Hessian defect sees thin tubes. These inputs do not imply a uniform John constant, a single connected bulk component, or a quantitative no-fjord/no-hole statement. A small-volume exceptional set can still disconnect the bulk or leave same-ray multiple crossings.

The final step also asserts that $H^1(D)$ -closeness of u_D to a radial quadratic implies L^1 -closeness of D to a ball by a coarea/trace argument. That implication needs quantitative nondegeneracy in a full collar around the zero level, not merely small surface measure of a low-gradient set on the level. In particular, it does not give a star-shaped graph.

Therefore the parenthetical line in `proof-step2.md` saying that “ H^2 defect plus volume exclusion gives a genuine star-shaped graph with small $H^{1/2}$ norm” is an assertion, not a proof. It is also not load-bearing once Chain B is removed.

4 The Step 4 graph passage assumes the missing input

`proof-step4.md` correctly says that if Step 2 produced a smooth closed level with pointwise Serrin control, nondegeneracy, and genuine star-shaped graph structure, then the implicit-function/overdetermined map would put it in a perturbative graph framework. But this is conditional on the missing Step-2 output:

$$\| |\nabla u| - c \|_{L^\infty(\Sigma_{\hat{t}})} \ll 1 \quad \text{and} \quad \Sigma_{\hat{t}} \text{ is a small graph.}$$

The corrected Step 1 gives a good level with an $O(\delta_T)$ removed layer and integrated $D_H + D_I$ control. It does not give this pointwise Serrin bound, a uniform gradient floor, a C^1 graph, or one radial crossing. Thus Step 4 is a valid downstream block, not a graph-entry proof.

5 Any-graph/Strategy B does not supply the reduction

`expA-anygraph-monotonicity.md` contains a useful warning: the naive radial homotopy monotonicity proof fails because higher-order shape terms have no sign on finite non-small graphs. It also points to a Pohozaev/ P -function route for star-shaped graph domains.

That route cannot be used as written for the present reduction. First, it assumes star-shaped graph geometry, which is exactly what we are trying to derive. Second, its advertised estimate

$$\delta_T(D) \gtrsim \int_D |D^2 u_D + I/n|^2$$

is the same unweighted Hessian coercivity contradicted by the high-frequency near-ball calculation above. Taking $\varepsilon \ll k^{-1}$ keeps $D_{\varepsilon,k}$ star-shaped with small C^1 norm, while the ratio of δ_T to the Hessian defect still tends to zero as $k \rightarrow \infty$.

6 What the $n = 2$ tentacle computation really gives

`tentacle-model.md` is genuinely useful. It shows that the specific “move inward to escape an outward thin tentacle” bookkeeping is favorable in dimension two. For a tube of width w and length L ,

$$\delta_T \simeq wL, \quad \text{escape layer} \simeq w^2,$$

so in $n = 2$ the layer is paid for when $L \gg w$. This removes one serious model obstruction.

But it is not a graph-entry theorem. It does not prove an arbitrary-geometry decomposition of every outward non-graph component of an analytic torsion level into chargeable fingers. It also says nothing by itself about inward fjords, holes, or same-ray multiple crossings.

7 What Plan 2 contributes

The Plan 2 radial slicing and metric-trace machinery is the strongest replacement for graph geometry in the repo. It gives:

1. finite-perimeter level identities with exact $D_H + D_I$;
2. same-center isoperimetric/radial trace control in L^1 -type metrics;
3. bad-ray multiplicity absorption once an annular bracket is known;
4. a plausible interval route at the correct $H^{1/2}$ scale.

This is not the same as graph entry. It allows a small set of bad rays and works in L^1 /metric distance, whereas the Plan 3 graph theorem needs a single-valued L^∞ radial graph or an $O(\delta_T)$ -cost repair to one.

Moreover the current Plan 2 interval route still records an explicit remaining conditional input. In `WIP_WeightedMetricTrace.tex` the bad-level kinetic hypothesis asks for

$$\int_{B_I} |\dot{F}_\rho|_{\mathcal{X}}^2 d\mu \leq C\delta_T, \quad \int_{B_{\tau_0}} |\dot{F}_\rho|_{\mathcal{X}}^2 d\rho \leq C\delta_T.$$

The patch note `PLAN2_INTERVAL_ROUTE_BAD_KINETIC_PATCH.md` closes the bad- I part and the slow-profile part on G_I , but leaves

$$\int_{B_{\tau_0} \cap B_I} |\dot{F}_\rho|_{\mathcal{X}}^2 d\rho \leq C\delta_T$$

open. Equivalently, with the crude deformation bound, it would suffice to prove

$$\int_{B_{\tau_0} \cap B_I} D_I(t(\rho)) d\rho \leq C\delta_T.$$

The existing identities control only the weighted integral $\int D_I(t(\rho))w(\rho) d\rho$, and on B_{τ_0} the weight $w = -t'(\rho)$ is small. Thus the interval route is promising, but it does not close the Plan 3 graph reduction.

8 What survives for the hybrid Plan 3 route

The following pieces are sound and should be retained.

1. The corrected Step 1 good-level extraction: one can stop at a regular near-boundary level with $O(\delta_T)$ removed volume and integrated $D_H + D_I$ control.
2. The superlevel torsion transfer and the final symmetric-difference transfer back to Ω .
3. The low-gradient surface estimate

$$\mathcal{H}^{n-1}(\{|\nabla u| \leq s\} \cap \Sigma_{\hat{t}}) \lesssim D_H(\hat{t}) s.$$

It is a real quantitative fact, but it is only measure control.

4. The direct radial-graph theorem in `H12_ASYMMETRY_DIRECT_PROOF.tex`: if a domain is a genuine small-amplitude radial graph, no slope or curvature bound is needed to get $\mathcal{A}^2 \lesssim \delta_T$.
5. The $n = 2$ outward-tentacle model: outward fingers should be chargeable at the correct $O(\delta_T)$ scale in dimension two.
6. The capacity identity in `PLANAR_RADIALIZATION_PROOF_ATTEMPT.tex`: if an inward missing set lies at positive torsion height in the filled comparison domain, then filling it has a torsion cost at least $a^2 \text{cap}(K)$.

These pieces strongly suggest the right missing theorem, but they do not prove it.

9 The exact remaining graph-reduction theorem

The honest target is the $O(\delta_T)$ radialization lemma:

Planar radialization lemma. Let $n = 2$, let $E = E_{\hat{t}}$ be the corrected Step-1 good level, and assume $\delta_T(\Omega)$ is small in the bounded normalized class. Then, after translation and rescaling, there is a set

$$G = \{re^{i\theta} : 0 < r < 1 + \varphi(\theta)\}, \quad \|\varphi\|_{L^\infty} \leq \eta_0,$$

such that

$$|E \Delta G| \leq C\delta_T(\Omega), \quad \delta_T(G) \leq C\delta_T(\Omega).$$

If this lemma is proved, the existing radial-graph theorem gives $\mathcal{A}(G)^2 \leq C\delta_T(G)$, the $O(\delta_T)$ repair gives $\mathcal{A}(E)^2 \leq C\delta_T(\Omega)$, and Step 5 transfers back to Ω .

The missing proof should split into three pieces:

1. an annular/core theorem putting the good level in a fixed thin annulus and giving positive torsion height on the pieces to be filled;
2. a planar capacity/modulus fill lemma for inward fjords, holes, and same-ray missing intervals;
3. an arbitrary-geometry outward pruning and bad-ray radialization lemma, using the $n = 2$ D_H -tentacle scaling plus Plan 2 bad-ray multiplicity.

10 Verdict

I think the memory is pointing to real material, but it closes less than the literal graph-reduction step. The repo already contains:

good level+metric/radial L^1 controls+ $n = 2$ outward-tentacle bookkeeping+capacity mechanism for fillable inward

What it does not contain is the theorem that turns these into either a genuine small L^∞ radial graph or an $O(\delta_T)$ -cost radialized replacement. The apparent closures in Plan 3 use a false Hessian inequality, assume the graph entry, or prove only model/metric statements.